

Task 1

Title	(SRS) – Library Management System
Project Team	- عمادالدين عصام محمد عبدالمعطي 2023040156 -
	- عبدالله صلاح سالم الشيخ 2023040138 -

Software Requirements Specification (SRS)

Library Management System (LMS)

Plan-Driven Model

1. Introduction

1.1 Purpose

The purpose of this document is to provide a complete, structured, and testable Software Requirements Specification (SRS) for the Library Management System (LMS). This document shall guide system design, implementation, verification, validation, and project management activities.

1.2 Intended Audience

- Software developers
- System architects
- Test engineers
- Project managers
- Stakeholders and clients

1.3 Intended Use

- Requirements validation
- System design and architecture
- Test case creation
- Progress tracking and change management

1.4 Product Scope

The LMS shall provide secure and efficient management of library resources such as books, borrowing and return operations, inventory management, overdue tracking, and reporting. The system shall support librarians, students, and administrators.

1.5 Definitions and Acronyms

LMS: Library Management System

SRS: Software Requirements Specification

FR: Functional Requirement

NFR: Non-Functional Requirement

2. Overall Description

2.1 User Needs

- Students require the ability to search for, borrow, and return books.
- Librarians require the ability to manage book inventory and user accounts.
- Administrators require high-level reports and analytical dashboards.
- All users require secure and responsive access to system services.

2.2 Assumptions and Dependencies

- All users must possess valid access credentials.
- The system depends on stable network connectivity to the shared database.
- The LMS shall operate on Windows and Linux environments.
- A relational database shall be used to store library data.

3. System Features and Requirements

3.1 Functional Requirements

3.1.1 Numbered Requirements

FR-1: The system shall allow users to search for books by title, author, or category.

FR-2: The system shall allow students to borrow available books.

FR-3: The system shall update book availability immediately after each transaction.

FR-4: The system shall allow users to return borrowed books.

FR-5: Librarians shall add, edit, and remove book records.

FR-6: Administrators shall generate monthly usage reports.

3.1.2 EARS Format Rules

When a student submits a borrow request, the system shall verify availability and issue the book.

When a due date is exceeded, the system shall generate an overdue notification.

When a librarian updates book information, the system shall save the updated metadata.

3.1.3 Specification by Example (BDD / Gherkin)

Example:

Given a student is logged in

And the book is available

When the student clicks 'Borrow'

Then the system shall mark the book as 'Borrowed'

And assign a due date automatically

3.2 Non- Functional Requirements

3.2.1 Performance

95% of all requests shall respond within 2 seconds.

3.2.2 Security

Only authenticated users can access borrowing, returning, and administrative functions.

3.2.3 Usability, Reliability, Compliance

- Usability: New users shall be able to perform a basic search within 10 minutes of training.
- Reliability: System uptime shall be at least 99% per month.
- Compliance: Password rules shall follow institutional IT security guidelines.

3.3 External Interface Requirements

3.3.1 Performance Requirements

The system shall support up to 200 concurrent users without performance degradation.

3.3.2 Safety Requirements

Critical data operations shall include rollback recovery in case of failure.

3.3.3 Security Requirements

User passwords shall never be stored in plain text.

3.3.4 Software Quality Attributes

The system shall maintain consistency and traceability of all requirements.

3.3.5 Business Rules

Borrowed books must be returned within 14 days unless renewed.

3.4 System Features

- Book Search Module
- Borrow/Return Module
- Inventory Management Module
- Overdue Tracking Module
- Reporting and Analytics Module

4. Other Requirements

4.1 Database Requirements

The system shall store all data in a relational SQL-based database.

4.2 Legal and Regulatory Requirements

The system shall comply with library data retention and privacy policies.

4.3 Internationalization and Localization

The system shall support English and Arabic interfaces.

4.4 Risk Management (FMEA Matrix)

Risks include data loss, downtime, unauthorized access. Mitigation plans shall be provided in a separate document.

5. Appendices

5.1 Glossary

A full list of definitions used in LMS.

5.2 Use Cases and Diagrams

Use Cases include: Search Book, Borrow Book, Return Book, Manage Inventory, Generate Reports.

5.3 TBD (To Be Determined) List

- Mobile app support
- AI-based book recommendations
- Offline mode functionality